

# Moulica Goli

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## Summary

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Software with 4+ years of experience designing and deploying 50+ distributed data pipelines and machine learning workflows using Apache Spark, Kafka and cloud native tools (GCP, AWS). Certified in AWS, Google Cloud Data Engineering and NVIDIA (Generative AI). Delivered ETL/ELT automation and Spark optimizations that reduced runtime by 40% and enabled processing of 5TB+ daily data. Experienced in fine-tuning large language models (LLMs) using PEFT and quantization techniques for production-scale AI applications.

## Education

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### Iowa State University, Ames

Jan 2024 – Dec 2025

Master of Science in Artificial Intelligence, Computer Science

Relevant Coursework - Artificial intelligence, Natural language processing, Machine learning, Deep learning, Design and analysis of Algorithms, Data Structures, Responsible AI

### International Institute of Information Technology, Bengaluru

Mar 2021 - Apr 2022

Post diploma in Software Development and Big Data

### JNTU Kakinada, Kakinada

Jun 2013 - May 2017

Bachelor of Technology in Computer Engineering

## Work Experience

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### Iowa State University - Graduate Research Assistant

Oct 2024 - Present

Ames, IA

- Engineered custom deep learning frameworks on HPC Nova infrastructure, boosting interpretation accuracy by 50%.
- Processed images with PIL and scikit-image to extract 25+ features, training Random Forest models that improved accuracy by 25% to enhance model transparency and align with Responsible AI practices.
- Explored multimodal AI by combining CLIP with fine-tuned LLM's through Hugging Face (Llama 3.2 multimodal), advancing research on image text alignment and improving inference quality in vision - language tasks.
- Researched and compared fine-tuning and activation steering vector techniques to optimize LLM inference, improving output accuracy while reducing token usage and context window size for more efficient deployment.
- Fine-tuned Large Language Models (LLMs) with project-specific data and applied Lora quantization techniques and deployed in the hugging hub with reduced computational requirements compared to other fine-tuning techniques.

### VERIZON - Software Engineer- 2

Aug 2021 - Dec 2023

Hyderabad, India

- Partnered with cross-functional teams to identify and source critical datasets, streamlining the data collection process and ensuring high-quality, timely data availability for model development.
- Performed in-depth data profiling and statistical analysis, applying advanced outlier detection and correlation analysis to ensure data integrity and build more reliable models.
- Engineered data preprocessing pipelines for missing values, feature scaling and imbalance correction (SMOTE) implemented PCA and correlation analysis on over 360 features to make important modeling choices.
- Conducted in-depth Explanatory Data Analysis (EDA) to generate data-driven operational insights, uncovering key trends that informed strategic business decisions.
- Tuned model hyperparameters on an imbalanced dataset by applying stratified k-fold cross-validation with GridSearchCV, reducing tuning time by 25% and identifying the optimal model configuration.

- Designed and trained a Linear Regression model to predict sales demand, using feature engineering and residual analysis, which made the predictions 15% more accurate.
- Developed and deployed ML models for sequence and optimization problems using Random Forest, XGBoost, k-Medoids clustering and TSP heuristics.
- Executed rigorous A/B hypothesis testing with income-based segmentation and uplift analysis, which boosted conversion rates by 40% and improved promotional spend efficiency by 10%.
- Introduced ML pipelines for training and scoring (Random Forest, XGBoost, SVM) and developed more than 30 using MLflow (MLOps) on GCP for feature selection, hyperparameter tuning.
- Containerized a Flask model scoring API using Docker and deployed it on Kubernetes, integrating CI/CD to automate model deployment and reduce manual effort by 40%.
- Practiced Agile methodology through daily stand-ups, sprint planning and Jira tracking, collaborating with upstream and downstream teams, which improved delivery timelines and reduced blockers.
- Deployed GraphQL-to-BigQuery ETL pipelines with built-in logging, alerting and YAML-based customization.
- Executed rigorous A/B hypothesis testing with income-based segmentation and uplift analysis, which boosted conversion rates by 40% and improved promotional spend efficiency by 10%.
- Practiced Agile methodology through daily stand-ups, sprint planning and Jira tracking, collaborating with upstream and downstream teams, which enhanced delivery timelines and reduced blockers.

## **Accenture - Application Development Analyst** **Hyderabad, India**

**Dec 2017 - Jul 2021**

- Reduced data curation time by 15 hours weekly by creating Hive views on external Hive tables.
- Enabled data integration with Azure Data Factory, making data available for reporting and analysis.
- Used Spark to modify data through UDF's and repartitioning which allowed for process information 65% faster.
- Leveraged Spark and Python to distribute data processing across large batch datasets, improving data ingestion and data processing speed by 65%, demonstrating expertise in scalable data solutions.
- Orchestrated data pipelines leveraging Hive and Oozie, resulting in 50% faster processing times for large-scale jobs and improved data quality delivered findings to fix the three biggest causes of data processing bottlenecks.
- Automated data pipelines in Oozie with Linux scripting and maintained version control through Git.
- Mentored 5 employees on clean code and best practices, conducted code reviews and led daily stand ups.

## **Projects**

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### **Multi-Agent Customer Support System | CrewAI, Multi-Agent Systems, Agentic AI**

- Built a multi-agent customer support framework using **CrewAI** to automate and streamline query handling.
- Designed specialized agents for query triage, response generation and escalation with guardrails.

### **RAG Project | LangChain, Weaviate, PyMuPDF, Google GenAI, Gemini LLM, Agentic AI**

- Implemented a Retrieval-Augmented Generation (RAG) pipeline with semantic search to query PDF's.
- Used PyMuPDF for text extraction, LangChain TextSplitter for chunking and Google Generative AI embeddings for vectorization.
- Stored embeddings in weaviate vector database and retrieved context for answer with Gemini 2.5 Flash LLM.

## **Technical Skills**

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**Languages:** Python, Scala, Scripting (Linux, Shell, Bash) , Java

**Cloud Technologies:** GCP, AWS

**Databases:** MySQL, Snowflake, NoSQL (HBase, Cassandra), PostgreSQL, Vector Database (Weaviate)

**Big Data:** Apache Spark, Hadoop, Hive, HDFS, Kafka, Airflow, Oozie

**ML/AI:** TensorFlow, PyTorch, Scikit-learn, pandas, Keras, LLM's, matplotlib, seaborn, RAG, langchain

**Visualization:** Tableau

## **Certifications**

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AWS Certified Data Engineer Associate | AWS Certified AI Practitioner | Google Cloud Professional Data Engineer |

